

Q1. What is Artificial Intelligence (AI)?

Answer:

Artificial Intelligence (AI) is a technology that helps computers do things that normally need human thinking. AI can:

Learn from data

Answer questions

Solve problems

Make decisions

Example:

ChatGPT answering questions

Google Maps showing directions

YouTube suggesting videos

Simple Definition:

AI is making a computer think and work like a human in some tasks.

Q2. Difference Between AI, Machine Learning, Generative AI, and Agentic AI

Answer:

Artificial Intelligence (AI)

AI is a technology that helps machines think and make decisions.

Example: Siri, Alexa

Machine Learning (ML)

Machine Learning is a part of AI. It helps computers learn from data and improve over time.

Example: Netflix movie recommendations

Generative AI

Generative AI creates new things like text, images, music, and videos.

Example: ChatGPT creating answers

Agentic AI

Agentic AI can plan and complete tasks by itself.

Example: An AI assistant that schedules meetings automatically

Simple Difference:

AI = Makes machines smart

ML = Helps machines learn

Generative AI = Creates new content

Agentic AI = Does tasks on its own

Q3. Explain Any Three Open-Source AI Models Available on Hugging Face

Answer:

1. Llama 3 Made by

Meta.

Uses:

Answers questions

Writes content

Helps with coding

2. Mistral 7B Made

by Mistral AI.

Uses:

Chatbots

Text generation

Quick responses

3. Falcon

Made by Technology Innovation Institute.

Uses:

Answering questions

Writing content

Learning and research

Simple Definition:

These are free AI models that can understand and generate text.

Q4. What is the Difference Between Traditional Programming and AI-Assisted Development?

Answer:

Traditional Programming AI-Assisted

Development

Human writes all code.

AI helps write code.

Takes more time.

Takes less time.

No AI support.

AI support is available.

More manual work.

Less manual work.

Example:

If you want a calculator program:

Traditional Programming: You write every line of code yourself.

AI-Assisted Development: You ask AI to create the code, and it helps you.

Simple Definition:

Traditional programming means writing everything yourself. AI-assisted development means AI helps you write the code.

Q5. Explain AI Hallucinations with One Example

Answer:

An AI hallucination happens when AI gives a wrong answer but says it as if it is correct.

Example:

A student asks:

"Who invented the smartphone?"

The AI may give a wrong person's name even though that person did not invent it.

This is called an AI hallucination.

How to Avoid It

Check important information from trusted sources.

Do not trust every AI answer without verification.

Simple Definition:

AI hallucination means AI gives false or made-up information and presents it as true.